Below is a 6-mark exam paper on "Database Management Systems (DBMS)" with short answer questions. The difficulty level is medium, and the total marks add up to 6. Answers are provided for each question.

Exam Paper: Database Management Systems (DBMS) (6 Marks)

Instructions:
- Answer all questions.
- Each question carries 6 marks.

Questions:

1. **What is a relational database? Explain with an example.** (2 marks)
   Answer:
   A relational database is a type of database that stores data in tables (relations) and defines relationships between tables using keys. It is based on the relational model introduced by E.F. Codd.
   Example: A database for a university may have tables like `Students`, `Courses`, and `Enrollments`, where `Enrollments` links `Students` and `Courses` using foreign keys.

2. **Define the term "primary key" with an example.** (1 mark)
   Answer:
   A primary key is a column (or set of columns) in a table that uniquely identifies each row. It cannot contain null values.
   Example: In a `Students` table, the `StudentID` column can be the primary key.

3. **What is normalization? Mention the first normal form (1NF) rule.** (1 mark)
   Answer:
   Normalization is the process of organizing data in a database to minimize redundancy and improve data integrity.
   1NF Rule: Each table cell must contain a single value (atomicity), and all entries in a column must be of the same data type.

4. **Explain the difference between SQL and NoSQL databases.** (2 marks)
   Answer:
   - **SQL (Relational) Databases:** Use structured tables, follow a schema, and support complex queries (e.g., MySQL, PostgreSQL).
   - **NoSQL (Non-relational) Databases:** Store unstructured or semi-structured data (e.g., JSON, key-value pairs) and are schema-less (e.g., MongoDB, Cassandra).